

ZeroDrift MK2: the complete build

Every part, every wire, every command, from an empty table to the rig locking onto the beacon on the website. Follow the sections in order. Every step ends with a **CHECK**. Don't go on until the check passes.

≈ 7 h

mechanics 3 h · wiring 2 h · bring-up 2 h

0.11° / step

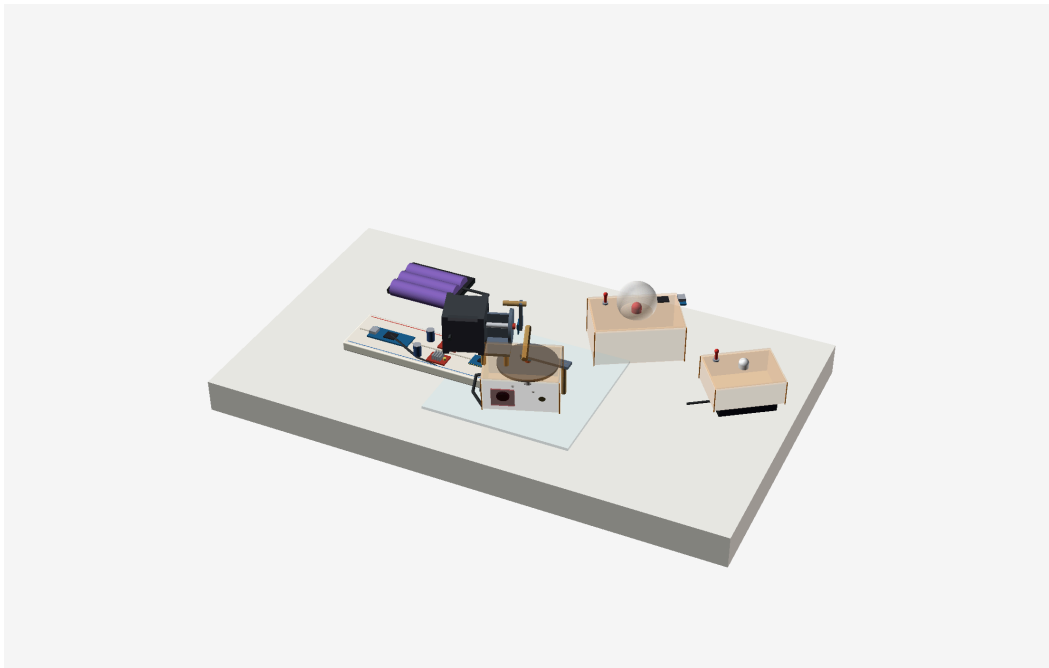
1/16 microstep, direct drive

0.088°

AS5600 encoder reading

±90° / ±18°

pan / tilt travel limit



What you are building: the MK2 gimbal (camera and laser in one head) tracking the 4 Hz red beacon, while ignoring the brighter white decoy that never blinks.

- 01 Parts: have / arriving / buy

02 Tools and software

03 Five safety rules

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05 Wiring: diagrams

06 Wiring: wire-by-wire list

07 Arduino: flash the firmware

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01 Parts

HAVE in hand **ARRIVING** ordered **BUY** get it today (Amar Robotics / hardware shop)

PART	QTY	STATUS	CHECK BEFORE YOU USE IT
☐ NEMA17 stepper, 1.8°/step	2	HAVE	4 wires each
☐ A4988 driver module	2	HAVE	Small resistors near the chip say R100 , R050 or R200 . Note it (for Vref, step 8.2)

☐	NEMA17 L-bracket	1–2	HAVE	Only the tilt motor needs one
☐	100 μ F 25 V capacitor	2	HAVE	Striped leg is –
☐	AS5600 magnetic encoder	2	HAVE	Solder the header pins if loose
☐	Diametric magnet 6 × 2.5 mm	2	HAVE	Usually comes with the AS5600. Must be diametric (test in step 4.4)
☐	TCA9548A I ² C multiplexer	1	ARRIVING	Solder the header pins
☐	USB webcam (UVC)	1	ARRIVING	Plug into the laptop and open it on the Live page before you strip the housing
☐	KY-008 laser module	1	HAVE	Pins marked S , middle, –
☐	Red filter (gel / red acrylic / sunglasses lens)	1	HAVE	Hold it over a phone camera: the room goes dim, a red LED stays bright
☐	ABS project box (head)	1	HAVE	
☐	Arduino Nano (rig) + Arduino Nano (beacon)	2	HAVE	Look at each USB port (Mini-USB or USB-C) and have the right cable
☐	12 V 3S Li-ion pack + barrel-to-screw-terminal adapter	1	HAVE	Charged: 11.1–12.6 V on the multimeter
☐	SPST toggle switch (motor kill switch)	1	HAVE	
☐	830-point breadboard + Dupont jumpers (M-M, M-F, F-F)	1	HAVE	
☐	3 mm acrylic, 6" × 6" (2 sheets)	2	HAVE	Sheet 1 = base. Sheet 2 = platform disc, tilt disc, encoder arms
☐	M3 screws, nuts, standoffs assortment	1	HAVE	Need: 4 standoffs $\approx$ 14–15 mm, and one brass standoff $\approx$ 40 mm (or two 20 mm joined)
☐	5 mm rigid flange coupling (shaft to plate)	2	BUY	Bore 5 mm with grub screws. Holds the platform and the tilt disc on the shafts with no play. $\approx$ ₹100–150 each
☐	3S (12.6 V) balance charger	1	BUY if not bought	Never charge a 3S pack on a single-cell charger (fire risk)
☐	USB-C to USB-A hub / adapter	1	BUY if laptop is USB-C only	The laptop needs 2 ports at once: Nano + webcam
☐	Beacon: 10 mm red LED, 220 Ω , switch, ping-pong ball, box, USB power bank	1 set	HAVE	Section 9
☐	Decoy: white LED, 68 Ω , 3×AA holder + cells, switch	1 set	HAVE	Section 9

Optional: a 100 mm lazy-Susan turntable bearing under the platform, only if the platform wobbles (step 4.3). A GT2 20T → 60T belt (3:1) gives 3× finer steps but costs a day of fitting. Skip it for Monday. The software already supports it (section 10, MK2 DRIVE).

02 Tools and software

Tools

- ☐ **Multimeter:** Vref, battery voltage, finding motor coil pairs. You can't build this without one.
- ☐ **Soldering iron** + solder: header pins on the TCA9548A and AS5600s
- ☐ Small screwdriver (plastic/ceramic ideal) for the A4988 pot; Phillips for M3
- ☐ Drill + 3 mm and 5–6 mm bits (a step bit if you have one); masking tape for acrylic
- ☐ Hacksaw / acrylic cutter, file or sandpaper
- ☐ Allen key for the flange-coupling grub screws
- ☐ Superglue (magnets), hot glue (camera, laser), double-sided foam tape, zip ties, insulation tape
- ☐ Wire stripper; ruler; marker; a printed protractor or phone level app
- ☐ Two old credit cards: stacked, they make a 1.5 mm gap gauge for the encoders

Software

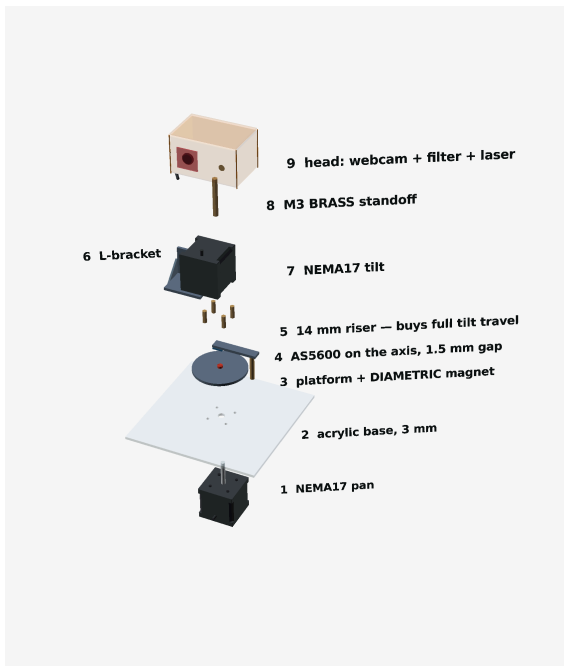
- ☐ **Arduino IDE 2** (arduino.cc/en/software)
- ☐ Library **AccelStepper** by Mike McCauley (Library Manager). `Wire` is built in.
- ☐ Firmware from the repo:
`tools/rig/rig_firmware_v2.ino` (rig) and
`tools/rig/beacon_firmware.ino` (beacon)
- ☐ **Chrome or Edge** (the website talks to the Nano over Web Serial; Safari/Firefox can't)
- ☐ Clone Nano (CH340 chip) not showing a port: install the CH340 driver

03 Five safety rules (each one kills a part)

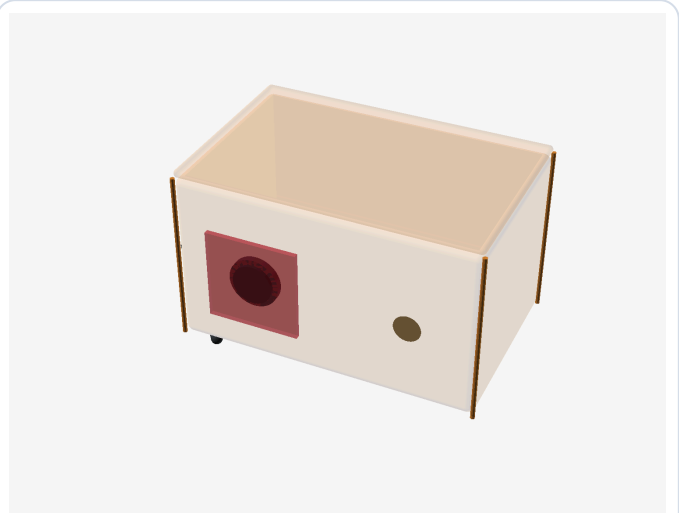
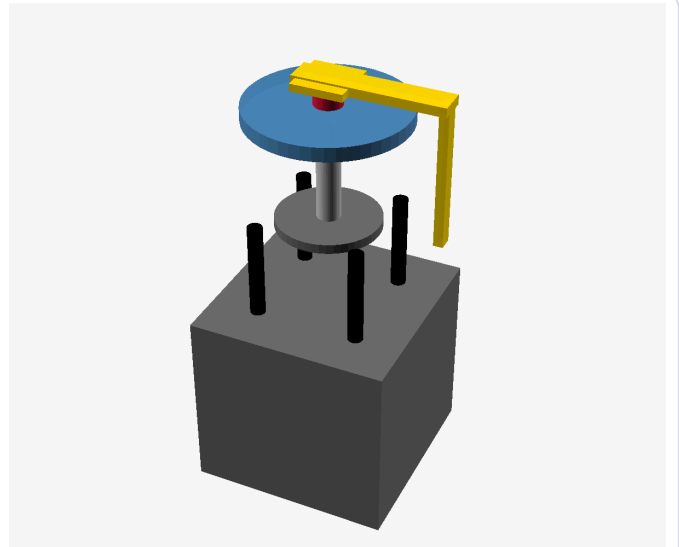
- 1 **Never plug or unplug a motor while the 12 V switch is on.** The A4988 dies instantly and silently.
- 2 **Capacitor on every VMOT**, 100 μ F, right at the driver pins, correct way round (stripe to GND). Without it the first power-on spike can kill the driver.
- 3 **One common ground.** Battery –, both drivers, the Nano and everything else share GND. Missing ground = motors buzz or do nothing.
- 4 **Laser:** never look into it and never point it at people. Test on a wall.
- 5 **12 V never goes to the Nano, the 5 V rail, VDD, the TCA or the AS5600s.** 12 V only touches VMOT. Check the rails with the multimeter before plugging boards in.

04 Mechanical build

The 3D step-by-step assembly shows every part below arriving in its place: click a stage, orbit, explode. Keep it open next to you.



Bottom to top: 1 pan motor · 2 base · 3 platform + magnet · 4 pan AS5600 · 5 risers · 6 L-bracket · 7 tilt motor · 8 brass standoff · 9 head.



- 4.1 Base plate (acrylic sheet 1).** Keep the protective film on, and put masking tape where you'll drill. Mark the centre. Hold the pan motor under the sheet, face up, and mark through its 4 screw holes (a 31 mm square). Drill the 4 holes at 3 mm and the centre at 6 mm. Drill slowly, no pressure, or acrylic cracks.

CHECK The motor shaft passes through the centre hole without touching the edge.

- 2 **4.2 Pan motor.** Bolt the motor **under** the base, shaft pointing up, with 4 M3×10 screws. Put 2 nuts on each screw between the motor and the plate as spacers: the motor's round boss (22 mm, 2 mm tall) must not press on the acrylic. Stand the base on 4 legs ≥ 55 mm tall (long standoffs, wood blocks or two books) so the motor hangs free. Put the motor cable out to the side you'll wire from.

CHECK The base sits flat and doesn't rock; the motor doesn't touch the table.

- 3 **4.3 Pan platform + magnet.** Cut a 60 mm disc from sheet 2. Draw 2 lines through its centre (a cross). Bolt it on top of a **5 mm flange coupling**, centred. Slide the coupling onto the pan shaft and leave a 2 mm gap to the base (the credit cards make a good spacer). Tighten the grub screw onto the flat of the shaft. Superglue a **diametric** magnet exactly on the centre of the cross.

CHECK Turn the motor shaft by hand: the disc spins flat, with no wobble and no rubbing. The magnet stays on the centre and doesn't draw a circle. If the disc wobbles, re-seat the coupling; last resort is the lazy-Susan bearing.

- 4 **4.4 Magnet test (2 minutes, saves hours).** Hold the second magnet near the glued one. A **diametric** magnet has its N and S on the *sides*: rolling one around the other flips attract/repel as you go round the edge. An axial magnet has N/S on the flat faces; it will read a constant and the encoder is dead with no error.

CHECK Both magnets are diametric.

- 5 **4.5 Pan encoder arm (does NOT turn).** Put an M3 post (stacked standoffs) on the base about 38 mm from the centre. Screw a strip of acrylic (≈ 12 × 50 mm) on top so it reaches over the centre. Fix the pan AS5600 under the arm, **chip face-down, chip exactly over the magnet**. Set the height with the 2 credit cards between chip and magnet (1.5 mm; 0.5–3 mm works). Use washers/nuts on the post to adjust.

CHECK Looking straight down, the chip is centred on the magnet within 0.5 mm, and the arm stays still when the platform turns.

- 6 **4.6 Risers + L-bracket.** Four M3 standoffs (14–15 mm) on the platform, L-bracket on top of them. The risers lift the bracket over the encoder arm. Check this in the 3D view. Put the bracket slightly off-centre so the tilt shaft ends up near the pan axis.

CHECK Turn the platform ±90° by hand: the bracket and risers pass over and around the encoder arm and post without touching.

- 7 **4.7 Tilt motor.** Bolt it to the L-bracket upright, shaft **horizontal**, at right angles to the way the head looks, so tilt nods up and down.

CHECK The shaft is level (phone level app).

- 8 **4.8 Tilt disc + magnet + encoder.** A 40 mm acrylic disc on the second flange coupling, on the tilt shaft. Glue the second diametric magnet at the centre of the disc's outer face. Fix the tilt AS5600 on a small arm bolted to the L-bracket (the side that does NOT turn with tilt). The chip faces the magnet, centred on the shaft line, 1.5 mm away.

CHECK Nodding the disc by hand, the gap stays at 1.5 mm and nothing rubs.

9 4.9 Brass standoff + head. Bolt the **brass** M3 standoff (≈ 40 mm) to the tilt disc near its edge, and hang the head box from it. Never use a printed or plastic part here: it sags and shakes. Inside the box, take the webcam board out of its housing. Hot-glue it behind a hole in the front, with the **red filter** over the lens (filter over the camera only, not the laser). Hot-glue the KY-008 laser beside it, about 20 mm away, pointing the same way (parallel).

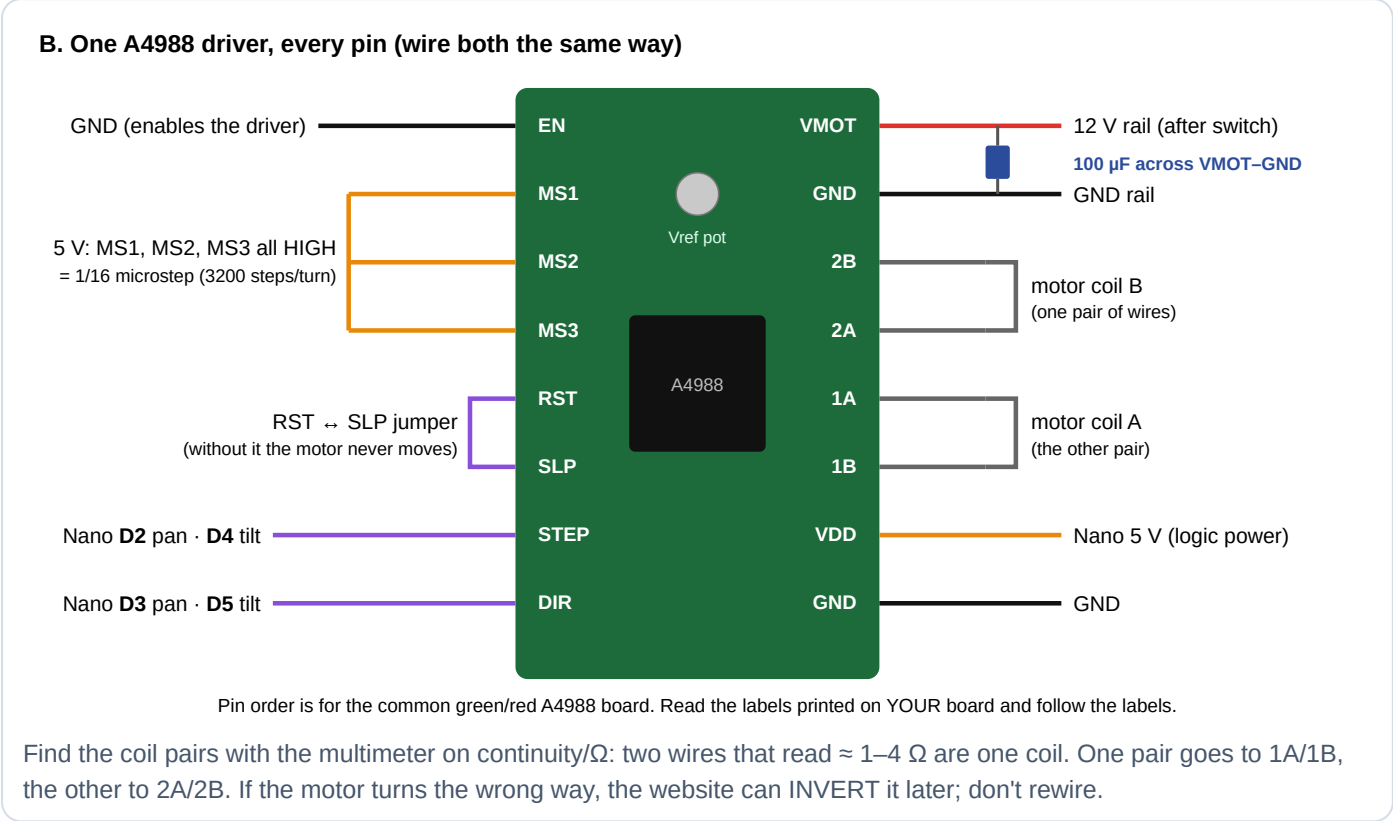
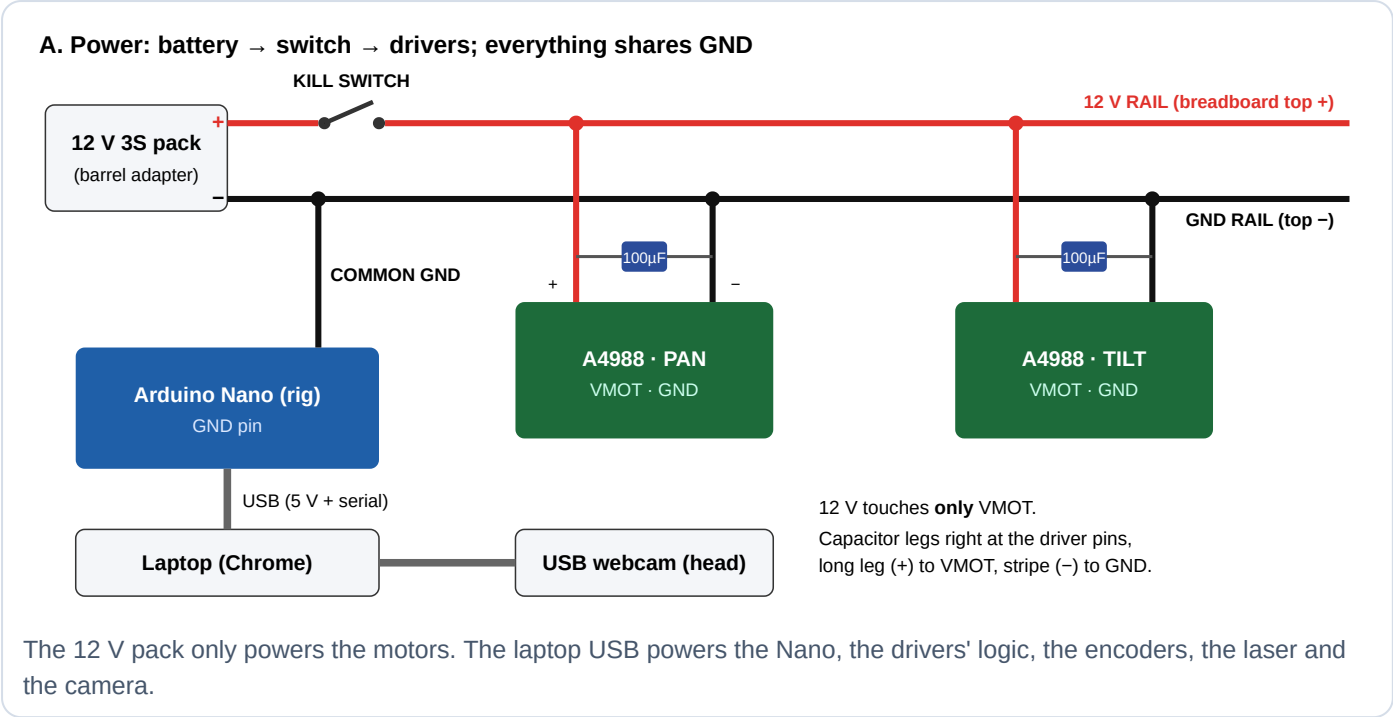
CHECK With the motors unpowered, move the head: it swings freely. Tilt $\pm 18^\circ$ clears the platform. Camera and laser point the same way (laser dot near the middle of the camera view on a wall 2 m away).

10 4.10 Cables. Run the camera USB, laser wires, tilt-motor cable and tilt-encoder wires down the bracket together (zip ties). Leave a loose **service loop** at the tilt joint and at the pan joint so full travel never pulls a wire. Keep the loop off the base plate.

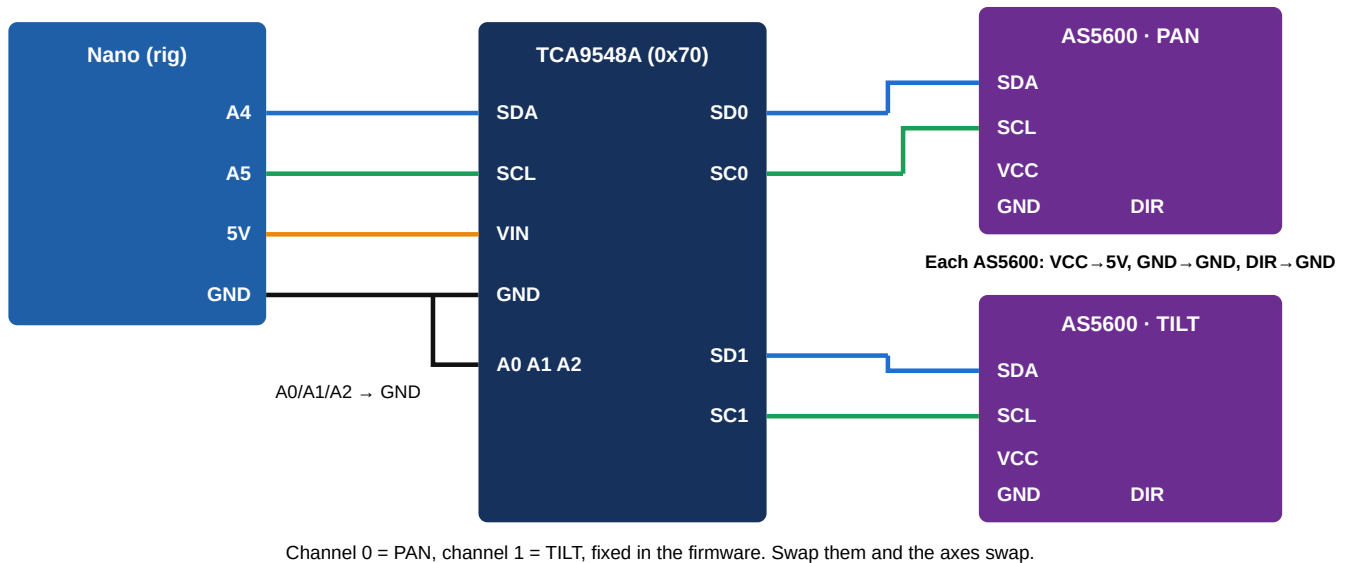
CHECK Pan $-90^\circ \rightarrow +90^\circ$ and tilt $-18^\circ \rightarrow +18^\circ$ by hand: no cable goes tight or catches.

05 Wiring: diagrams

Wire colours used below: **red = 12 V**, **orange = 5 V**, **black = GND**, **blue = SDA**, **green = SCL**, **purple = signals**. Do all wiring with the battery **unplugged** and the Nano USB **unplugged**.

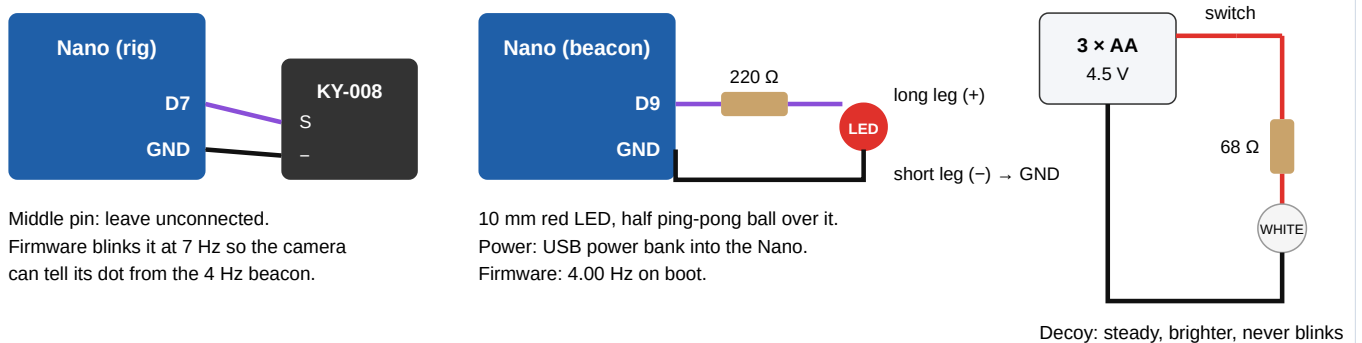


C. Encoders: Nano → TCA9548A → two AS5600 (both are address 0x36, so the mux is required)



Tie DIR to GND (a floating DIR makes the reading jump). If an AS5600 board has a closed "3V3" solder jumper, power that board from the Nano's 3V3 pin instead of 5 V. On the rig, the tilt encoder's 5 wires ride the pan axis: give them a service loop.

D. Laser (on the rig) · beacon (second Nano) · decoy (no Nano)



The beacon Nano can drive the LED straight from D9 through 220 Ω (≈ 14 mA). The 2N2222 in the kit is not needed.

06 Wiring: wire-by-wire list (tick each one)

Breadboard layout: top rail pair = **12 V / GND**, bottom rail pair = **5 V / GND**. Nano in the middle, the two A4988s either side, TCA9548A at one end. Leave the 12 V rail's + side empty until step 8.3.

#	FROM	TO	NOTE
Power and ground			
☐ 1	Battery + (barrel adapter)	Kill switch, pin 1	Red wire
☐ 2	Kill switch, pin 2	Top rail +, the 12 V rail	
☐ 3	Battery –	Top rail –, the GND rail	
☐ 4	Top GND rail	Bottom GND rail	Common ground. Don't skip it.
☐ 5	Nano GND	Bottom GND rail	
☐ 6	Nano 5V	Bottom + rail, the 5 V rail	Never connect this to the 12 V rail
PAN driver (A4988 #1)			
☐ 7	12 V rail	PAN VMOT	
☐ 8	GND rail	PAN GND (next to VMOT)	
☐ 9	100 µF: long leg / stripe leg	PAN VMOT / PAN GND	Right at the pins
☐ 10	5 V rail	PAN VDD	
☐ 11	GND rail	PAN GND (next to VDD)	
☐ 12	5 V rail	PAN MS1, MS2, MS3	3 short jumpers → 1/16 step
☐ 13	PAN RST	PAN SLP	Short jumper between the two pins
☐ 14	PAN EN	GND rail	
☐ 15	Nano D2	PAN STEP	
☐ 16	Nano D3	PAN DIR	
☐ 17	Pan motor coil A / coil B	PAN 1A-1B / 2A-2B	Pairs from the multimeter test
TILT driver (A4988 #2): same as PAN, different Nano pins			
☐ 18–25	Repeat wires 7–14 on the TILT driver		VMOT, GND, cap, VDD, GND, MS1-3, RST-SLP, EN
☐ 26	Nano D4	TILT STEP	
☐ 27	Nano D5	TILT DIR	
☐ 28	Tilt motor coil A / coil B	TILT 1A-1B / 2A-2B	
Encoders			
☐ 29	Nano A4	TCA9548A SDA	Blue

☐	30	Nano A5	TCA9548A SCL	Green
☐	31	5 V rail / GND rail	TCA9548A VIN / GND	
☐	32	GND rail	TCA9548A A0, A1, A2	Address 0x70
☐	33	TCA SD0 / SC0	PAN AS5600 SDA / SCL	Channel 0 = pan
☐	34	5 V rail / GND rail / GND rail	PAN AS5600 VCC / GND / DIR	
☐	35	TCA SD1 / SC1	TILT AS5600 SDA / SCL	Channel 1 = tilt
☐	36	5 V rail / GND rail / GND rail	TILT AS5600 VCC / GND / DIR	
Laser and USB				
☐	37	Nano D7	KY-008 S	
☐	38	GND rail	KY-008 –	Middle pin: nothing
☐	39	Nano USB	Laptop	Serial + 5 V
☐	40	Webcam USB	Laptop (hub if needed)	

Not used in this build: D8 (vibration injector) and D9/D10 (old servo fine stage). Leave them empty.

07 Arduino: flash the firmware

1 Arduino IDE → **Tools** → **Manage Libraries** → search `AccelStepper` → install (Mike McCauley).

2 Open `tools/rig/rig_firmware_v2.ino`. Direct drive needs no edits. (Belt 20T → 60T only: set `PAN_GEAR_RATIO` / `TILT_GEAR_RATIO` to `3.0`.)

3 **Tools** → **Board** → **Arduino Nano. Processor** → **ATmega328P**. If upload fails with "not in sync", choose **ATmega328P (Old Bootloader)** (most clones).

4 **Tools** → **Port** → the Nano (appears when you plug it in; `usbserial` / `wchusbserial` on Mac, `COMx` on Windows). Click **Upload**.

5 **Tools** → **Serial Monitor**. Set **115200 baud** and line ending **Newline**.

CHECK It prints `# ZeroDrift Mk2 ready, encoders=yes`. "encoders=no" → section 12.

08 First power-on and tests

- 1 8.1 Rails check (battery unplugged, Nano on USB).** Multimeter DC V: 5 V rail to GND $\approx$ 4.7–5.0 V. The 12 V rail reads 0.

CHECK 5 V present; nothing is hot.

- 2 8.2 Set Vref on both drivers (Nano USB only, battery OFF).** Black probe on GND, red probe touching the metal of the driver's little pot. Turn gently to read:

RESISTOR MARKING	R100 (MOST COMMON)	R050	R200
Vref ($\approx$ 0.7 A per coil)	0.55 V	0.28 V	1.1 V

CHECK Both drivers set. (Vref = current $\times$ 8 $\times$ resistor. The motors hold firmly, and the drivers only get warm.)

- 3 8.3 Motors on.** Motors plugged in. Flip the kill switch ON.

CHECK Both shafts become stiff (holding). No smell, no hot chip after 1 minute. If a driver gets too hot to touch, lower Vref 10 %.

- 4 8.4 Serial tests.** Type each line into the Serial Monitor and press Enter:

you type	what should happen
?	S 0 0 E 0 E = encoder reading (C = no encoders)
P267 T0	pan turns 30° (8.889 steps per degree)
?	S 26x 0 E 0 pan $\approx$ 267 $\pm$ 3 from the encoder
P0 T0	pan returns
T89	tilt nods 10°
T0	tilt returns
L1	laser ON (steady)
L7.0	laser blinks 7 Hz
L0	laser OFF
!	# encoder reads 10.0x deg (expect ~10.00 ...)
Z	# zeroed here = straight ahead

CHECK Pan and tilt both move, both come back, and the encoder numbers match the command within a few steps.

- 5 8.5 Encoder direction.** After P267 T0, ? must say about +267. If it says -267, move that AS5600's DIR wire from GND to 5 V. Do the same for tilt with T89.

CHECK Encoder sign equals the command sign on both axes.

- 6 8.6 Close the Serial Monitor,** or the website can't open the port.

09 Beacon and decoy

- 1 Beacon.** Second Nano: D9 → 220 Ω → red LED long leg; short leg → GND. LED through a hole in the box lid, half a sanded ping-pong ball glued over it. Flash `tools/rig/beacon_firmware.ino`. Power from a USB power bank.
CHECK Serial Monitor (115200) shows `ZeroDrift beacon ready F4.00 B255 M1`. The LED blinks 4 times a second. Commands: `F6.0` sets 6 Hz, `M0` steady, `M1` blink.
- 2 Decoy.** 3×AA → switch → 68 Ω → white LED → back to the battery –. Steady, brighter than the beacon.
CHECK On the Live page the decoy never gets locked, even when it's the brightest thing in view.

10 Connect to the website

Open zerodrift-fsoc-pat.netlify.app/live in Chrome or Edge.

- 1 START CAMERA**, then **MK2 CAMERA (USB)**. The view switches to the head's camera (HUD says *MK2 CAM*).
CHECK Wave in front of the head: you see it on screen.
- 2 CONNECT RIG** → pick the Nano's port.
CHECK The RIG card says **MK2 · encoders**.
- 3 MK2 DRIVE = Direct drive** (or 3:1 only if you fitted belts AND set the firmware ratio).
- 4 TEST MOTION:** it should pan right, tilt up, then blink the laser.
CHECK Wrong direction → tick **INVERT PAN / INVERT TILT**. Wrong axis → **SWAP PAN / TILT**. Repeat until correct.
- 5 BEACON RATE = 4 Hz** (the beacon's rate). Only that rate (±10 %) is locked; a 3 Hz or 5 Hz light is ignored.
- 6** Beacon on, 1–3 m away.
CHECK It locks (green box). The head turns until the beacon is in the centre. The GIMBAL readout ends in **enc** (encoder confirmed).
- 7** Bright room: the red filter over the camera is what makes this work. If the room is still too bright, add a second layer or dim the lights. The **EXPOSURE** slider helps where the browser supports it (Windows/Android).

11 Calibrate (15 minutes)

- 1 Straight ahead = zero.** Kill switch OFF, point the head straight ahead by hand, then press **CONNECT RIG**. Connecting restarts the Nano, and it takes the current position as zero. Then switch the motors ON. (In the Serial Monitor, `Z` does the same.)

2 45° check (Serial Monitor, before connecting the website). Send `P400 T0` (45°) and check it with a printed protractor under the platform.

CHECK $45^\circ \pm 1^\circ$, and `?` reads ≈ 400 . If it's off by exactly $2\times$, $4\times$ or $\frac{1}{2}$, the MS1-3 jumpers are wrong.

3 SET AIM (the important one). Put a sheet of white paper behind the beacon. When the beacon is locked, the laser switches on by itself (blinking at 7 Hz). Click **SET AIM**, then click exactly on the laser dot in the video.

CHECK From now on the rig puts the beacon on the laser dot, not the image centre. This cancels the camera–laser offset (parallax).

4 Balance. Motors off, leave the head at $+15^\circ$ and at -15° . If it drops, move the camera/laser/standoff until it stays. A balanced head doesn't sag or lose steps.

12 Troubleshooting

SYMPTOM	MOST LIKELY CAUSE	FIX
Motor doesn't move, no holding	RST–SLP jumper missing; no 12 V; EN not to GND	Wires 13/14; switch on; meter VMOT ≈ 12 V
Motor buzzes / shakes, doesn't turn	Coil pairs mixed (1A with 2A)	Re-find pairs with the meter; one pair on 1A/1B, the other on 2A/2B
Moves but angle is $2\times/4\times/\frac{1}{2}$ off	MS1-3 not all at 5 V	Wire 12 (all three HIGH = 1/16)
Misses steps / loses position	Vref too low, unbalanced head, cable pulling	Vref +10 %, balance the head (11.4), free the service loop
Driver very hot / shuts off	Vref too high	Lower Vref; stick a small heatsink on the chip
<code>encoders=no</code>	SDA/SCL swapped; TCA not powered; A0-A2 floating; headers not soldered	A4 $\rightarrow$ SDA, A5 $\rightarrow$ SCL; 5 V on VIN; A0-A2 to GND; solder headers
<code>?</code> encoder number doesn't change	Axial magnet, or chip off-axis / gap too big	Magnet test (4.4); centre chip; gap 1–2 mm
Encoder sign opposite to command	AS5600 DIR	Move DIR from GND to 5 V (8.5)
Upload fails "not in sync"	Clone bootloader	Processor $\rightarrow$ ATmega328P (Old Bootloader)
CONNECT RIG: port busy / nothing	Serial Monitor still open; Safari	Close Serial Monitor; use Chrome/Edge
MK2 CAMERA shows the laptop camera	Webcam not plugged / needs permission	Plug it in, reload, allow camera, press MK2 CAMERA (USB) again
Won't lock in a bright room	Camera over-exposed	Red filter over the lens (2 layers if needed), dim lights
Locks, then turns away from the beacon	Axis direction	INVERT / SWAP in the RIG card, TEST MOTION again
Beacon on the centre, not on the laser dot	Aim not set	SET AIM (11.3)
Wrong rate shown / not locking	BEACON RATE $\neq$ beacon	Set BEACON RATE to the beacon's F value (4 Hz default)

Quick reference

NANO PIN	GOES TO	NANO PIN	GOES TO
D2 / D3	PAN STEP / DIR	A4 / A5	TCA9548A SDA / SCL
D4 / D5	TILT STEP / DIR	5V	VDD ×2, MS1-3 ×2, TCA VIN, AS5600 VCC ×2
D7	Laser S	GND	Common ground (battery –, drivers, TCA, AS5600, laser, EN, A0-A2, DIR)

ZeroDrift · Team ZeroDrift, Jabalpur Engineering College · SIH 2026 PS26169. Firmware: tools/rig/rig_firmware_v2.ino. Geometry: docs/cad (same source as the 3D assembly).